Year 7 Mathematics – Practice Exams 1 to 4

Answers and marking guide

Award marks for correct working as well as correct answers. In Part B, Questions 1 and 2, one mark is for the working (with a reason, e.g. ‘angle sum of a triangle is 180°’) and one for the answer. In Part C, Question 12 i, award 2 marks for a dot plot with a labelled, evenly-spaced number line and the correct number of dots above each value (1 mark if there are small errors).

Skills checklist coverage (the same in every practice exam)

#	Skill (from the Year 7 skills checklist)	Practice exam question
1	Know the meaning of mean, median, mode and range	A1, A2
2	Know the order of operations	A3
3	Understand ascending and descending order	A4
4	Name the place value of decimal digits	A5
5	Understand index notation	A6
6	Know what a variable (pronumeral) is	A7
7	Know that the angles in a triangle add to 180°	A8
8	Identify the features of an isosceles triangle	A9
9	Know what perimeter measures	A10
10	Understand negative numbers on a number line	A11
11	Know the sample space for rolling a die and what P(E) means	A12, A13
12	Classify angles (acute, right, obtuse, straight, reflex)	A14
13	Know the angle relationships on parallel lines	A15
14	Find a missing angle in a triangle using the angle sum	B1
15	Find the exterior angle of a triangle	B2
16	Identify equivalent fractions	B3
17	Multiply decimals and fractions without a calculator	B4
18	Add and subtract integers (negative numbers)	B5 a, b
19	Use the order of operations with brackets and integers	B5 c
20	Convert between improper fractions and mixed numbers	B6
21	Write an algebraic expression from a worded statement	B7
22	Substitute values into algebraic expressions	B8
23	Simplify expressions by collecting like terms	B9 i–iii
24	Multiply algebraic terms, e.g. 3 × 3y	B9 iv
25	Use mental strategies for addition and multiplication	B10
26	Solve one-step equations (+, −, ×, ÷)	B11 i–iv
27	Solve two-step equations, including ones with a fraction	B11 v, vi
28	Construct a dot plot from a list of data	C12 i
29	Read a dot plot to find the total and count values ‘less than’ a number	C12 ii, v
30	Find the mode and range of a data set	C12 iii, iv
31	Identify an outlier in a data set	C12 vi
32	Calculate the mean of a data set	C13 i
33	Find the median of a data set	C13 ii
34	Work out the new mean and median when a value is added	C13 iii, iv
35	Read and interpret a column graph, including finding totals	C14 i–iii
36	Write a value from a graph as a fraction of a whole	C14 iv
37	List the sample space for a spinner	C15 a
38	Calculate the probability of a simple event	C15 b
39	Find the complement of an event	C15 c

Practice Exam 1 – Answers

Part	Q	Answer
A	1	c) Mode
A	2	b) 10
A	3	c) 2 + 4
A	4	b) 15, 11, 8, 3
A	5	b) Hundredths
A	6	c) 5 × 5 × 5 × 5
A	7	b) A variable – a letter that stands for a number
A	8	b) 180°
A	9	c) Isosceles
A	10	c) 16 cm
A	11	a) −9
A	12	c) {1, 2, 3, 4, 5, 6}
A	13	b) The probability of event E happening
A	14	c) Obtuse
A	15	a) Equal
B	1	x = 180° − 48° − 77° = 55°
B	2	a = 55° + 65° = 120° (or 180° − 60° = 120°)
B	3	a) True b) False
B	4	a) 0.8 b) $\frac{2}{15}$
B	5	a) 5 b) −13 c) 12
B	6	a) $2\frac{1}{3}$ b) $\frac{17}{5}$
B	7	i) $x+5$ ii) $x \div 4$ or $\frac{x}{4}$
B	8	i) 13 ii) 2 × 6 − 4 = 8
B	9	i) $9m$ ii) $8gh$ iii) $5k+5$ iv) $8w$
B	10	i) 40 (13 + 7 = 20, 8 + 12 = 20) ii) 1800 iii) 95 (20 × 5 − 5)
B	11	i) m = 8 ii) h = 13 iii) y = 7 iv) n = 20 v) x = 5 vi) w = 7
C	12	i) Dot plot with a labelled number line from 1 to 10 – number of dots: 1: 3, 2: 7, 3: 8, 4: 4, 5: 1, 10: 1 ii) 24 iii) Mode = 3 iv) Range = 10 − 1 = 9 v) 10 vi) Yes – 10 is an outlier (it is far away from the rest of the data)
C	13	i) Mean = 54 ÷ 9 = 6 ii) Median = 6 (in order: 2, 3, 4, 5, 6, 7, 8, 9, 10) iii) New median = 6.5 iv) New mean = 68 ÷ 10 = 6.8
C	14	i) 4 hours ii) Saturday iii) 2 + 1 + 3 + 2 + 4 + 1 + 6 = 19 hours iv) $\frac{6}{24} = \frac{1}{4}$
C	15	a) {red, blue, green, yellow} b) $\frac{3}{8}$ c) Not blue (spinning red, green or yellow); P(not blue) = 1 − P(blue) = $\frac{6}{8} = \frac{3}{4}$

Total: 63 marks (Part A 15, Part B 30, Part C 18)

Practice Exam 2 – Answers

Part	Q	Answer
A	1	d) Median
A	2	b) 6
A	3	b) 4
A	4	b) Smallest to largest
A	5	c) 2
A	6	a) 2^6
A	7	a) Pronumeral
A	8	b) 50°
A	9	a) Isosceles
A	10	b) Skirting board to go around the edge of a room
A	11	b) −5, −1, 0, 3
A	12	c) 6
A	13	b) Impossible
A	14	c) Reflex
A	15	b) Add up to 180°
B	1	x = 180° − 90° − 34° = 56°
B	2	a = 40° + 75° = 115° (or 180° − 65° = 115°)
B	3	a) True b) False ($\frac{5}{8} = \frac{20}{32}$)
B	4	a) 1.2 b) $\frac{3}{8}$
B	5	a) −5 b) −12 c) 15
B	6	a) $2\frac{3}{4}$ b) $\frac{11}{2}$
B	7	i) $x + 8$ ii) $6x$
B	8	i) 8 ii) 2 × 3 + 5 = 11
B	9	i) $12a$ ii) $6pq$ iii) $5t + 6$ iv) $20n$
B	10	i) 40 (15 + 5 = 20, 8 + 12 = 20) ii) 4000 iii) 126 (20 × 6 + 6)
B	11	i) k = 5 ii) p = 14 iii) c = 8 iv) m = 18 v) n = 7 vi) z = 11
C	12	i) Dot plot with a labelled number line from 1 to 8 – number of dots: 1: 8, 2: 7, 3: 3, 4: 1, 8: 1 ii) 20 iii) Mode = 1 iv) Range = 8 − 1 = 7 v) 15 vi) Yes – 8 is an outlier (it is far away from the rest of the data)
C	13	i) Mean = 45 ÷ 9 = 5 ii) Median = 5 (in order: 1, 2, 3, 4, 5, 6, 7, 8, 9) iii) New median = 5.5 iv) New mean = 60 ÷ 10 = 6
C	14	i) 3 hours ii) Tuesday iii) 6 + 2 + 1 + 3 + 2 + 4 + 5 = 23 hours iv) $\frac{6}{24} = \frac{1}{4}$
C	15	a) {red, blue, green, yellow} b) $\frac{4}{10} = \frac{2}{5}$ c) Not red (spinning blue, green or yellow); P(not red) = 1 − P(red) = $\frac{7}{10}$

Total: 63 marks (Part A 15, Part B 30, Part C 18)

Practice Exam 3 – Answers

Part	Q	Answer
A	1	c) Range
A	2	b) 5
A	3	b) 2 + 3 × 4 = 14
A	4	a) 5, 0.55, 0.5, 0.05
A	5	c) Thousandths
A	6	b) 27
A	7	c) y + 4
A	8	b) 90°, 45°, 45°
A	9	b) It has two equal sides and two equal angles
A	10	b) 24 cm
A	11	c) 2 °C
A	12	b) 3
A	13	d) $\frac{3}{2}$
A	14	b) Right
A	15	c) Alternate
B	1	x = 180° − 112° − 31° = 37°
B	2	a = 50° + 85° = 135° (or 180° − 45° = 135°)
B	3	a) True b) False ($\frac{2}{9} = \frac{6}{27}$)
B	4	a) 1.8 b) $\frac{3}{20}$
B	5	a) −6 b) −10 c) 15
B	6	a) $2\frac{3}{5}$ b) $\frac{19}{6}$
B	7	i) $x - 7$ ii) $2x + 1$
B	8	i) 13 ii) 3 × 2 + 5 = 11
B	9	i) $13b$ ii) $9mn$ iii) $2r + 10$ iv) $12z$
B	10	i) 50 (24 + 6 = 30, 17 + 3 = 20) ii) 5400 iii) 124 (30 × 4 + 4)
B	11	i) a = 8 ii) b = 14 iii) c = 9 iv) e = 12 v) y = 5 vi) q = 13
C	12	i) Dot plot with a labelled number line from 2 to 14 – number of dots: 2: 1, 3: 3, 4: 5, 5: 6, 6: 3, 7: 1, 14: 1 ii) 20 iii) Mode = 5 iv) Range = 14 − 2 = 12 v) 4 vi) Yes – 14 is an outlier (it is far away from the rest of the data)
C	13	i) Mean = 63 ÷ 9 = 7 ii) Median = 7 (in order: 3, 4, 5, 6, 7, 8, 9, 10, 11) iii) New median = 7.5 iv) New mean = 80 ÷ 10 = 8
C	14	i) 5 hours ii) Thursday iii) 4 + 2 + 3 + 5 + 1 + 3 + 9 = 27 hours iv) $\frac{9}{24} = \frac{3}{8}$
C	15	a) {red, blue, green} b) $\frac{2}{6} = \frac{1}{3}$ c) Not green (spinning red or blue); P(not green) = 1 − P(green) = $\frac{3}{6} = \frac{1}{2}$

Total: 63 marks (Part A 15, Part B 30, Part C 18)

Practice Exam 4 – Answers

Part	Q	Answer
A	1	b) Mean
A	2	a) 3
A	3	b) 18
A	4	b) 7, 4, 0, −2
A	5	c) 8.31
A	6	b) 6^3
A	7	b) A letter or symbol that stands for a number
A	8	b) 55°
A	9	c) 70°
A	10	c) Carpet to cover a bedroom floor
A	11	b) −4
A	12	d) 7
A	13	d) Certain
A	14	c) Straight
A	15	c) Corresponding
B	1	x = 180° − 64° − 71° = 45°
B	2	a = 62° + 48° = 110° (or 180° − 70° = 110°)
B	3	a) True b) False ($\frac{7}{10} = \frac{28}{40}$)
B	4	a) 3.5 b) $\frac{2}{21}$
B	5	a) 7 b) −15 c) 12
B	6	a) $2\frac{5}{6}$ b) $\frac{17}{3}$
B	7	i) $9x$ ii) $12 - x$
B	8	i) 15 ii) 3 × 4 + 2 = 14
B	9	i) $13c$ ii) $7xy$ iii) $5n + 9$ iv) $14k$
B	10	i) 50 (18 + 2 = 20, 7 + 23 = 30) ii) 2800 iii) 87 (30 × 3 − 3)
B	11	i) v = 7 ii) w = 15 iii) x = 6 iv) y = 21 v) z = 7 vi) t = 10
C	12	i) Dot plot with a labelled number line from 0 to 9 – number of dots: 0: 4, 1: 8, 2: 6, 3: 2, 4: 1, 9: 1 ii) 22 iii) Mode = 1 iv) Range = 9 − 0 = 9 v) 12 vi) Yes – 9 is an outlier (it is far away from the rest of the data)
C	13	i) Mean = 81 ÷ 9 = 9 ii) Median = 9 (in order: 5, 6, 7, 8, 9, 10, 11, 12, 13) iii) New median = 9.5 iv) New mean = 100 ÷ 10 = 10
C	14	i) 1 hours ii) Friday iii) 4 + 2 + 3 + 1 + 2 + 6 + 3 = 21 hours iv) $\frac{4}{24} = \frac{1}{6}$
C	15	a) {red, blue, green, yellow} b) $\frac{3}{12} = \frac{1}{4}$ c) Not green (spinning red, blue or yellow); P(not green) = 1 − P(green) = $\frac{8}{12} = \frac{2}{3}$

Total: 63 marks (Part A 15, Part B 30, Part C 18)